

EDUCATION

Program	Institution	CGPA/%	Year
M.Tech Data Science + B.Tech Engineering Design	IIT Madras	9.26/10	2028
Class XII (CBSE)	Babaji Vidhyashram, Chennai	96.4%	2022–23
Class X (CBSE)	Babaji Vidhyashram, Chennai	97.4%	2020–21

SCHOLASTIC ACHIEVEMENTS

- Earned the IIT Madras **Young Research Fellowship** (top 25 of 300+) for deep learning in computational geometry.
- Received an **INR 82k** Global Engagement travel grant as 1 of 2 IITM undergraduates attending AI conferences.
- Ranked **3rd of 78** after sophomore year and earned **9.56 SGPA** in the first semester of M.Tech Data Science.

PUBLICATIONS

- **ConvPSNet: A convolution-based single-step approach for pruned skeleton extraction** *Hong Kong SIGGRAPH Asia 2025, Technical Communications*
Bincy Antony M, Anshuman Mishra et al.
 - Developed a single-step convolutional framework to extract pruned skeletons from noisy 2D shape samples.
 - Curated synthetic shape datasets spanning diverse geometries and boundary perturbations for model training.
 - Designed scalable augmentation pipelines to generate varied noisy shape configurations across diverse scenarios.
- **DistillSkel: Learning pruned skeletons of 2D shapes through progressive knowledge distillation** *Istanbul, Türkiye Shape Modelling International 2026, Technical Paper*
Bincy Antony M, Anshuman Mishra et al.
 - Distilled a U-Net from **31M to 31K parameters (1000× reduction)** with **<5% F1 loss** in skeletonization accuracy.
 - Introduced a novel junction-preservation loss to retain skeleton connectivity under shape-boundary noise.
 - Led in-house creation of a **2k+ shape** dataset through augmentation, preprocessing and guided synthesis.

RESEARCH EXPERIENCE

- **Young Research Fellow | IIT Madras** *Jul '25 – Jul '26*
 - Benchmarked **10+** skeletonization approaches while investigating deep-learning-based shape skeletonization.
 - Modeled midcurves and midsurfaces with signed distance fields under Prof. Ramanathan Muthuganapathy.
 - Created **35+** figures, attended SIGGRAPH Asia 2025 and presented an industry poster at IISc Bangalore.
- **Project Member - Team Suncast | AI Club × Horizon Club** *May '24 – May '25*
 - Combined GATs and LSTMs to model spatial–temporal dependencies in correlation-based solar-flare graphs.
 - Improved classification accuracy by **>8%** over non-graph baselines through joint spatial–temporal learning.
 - Presented research results to over **5k+** attendees across CFI Research Conclave '24 and CFI Open House '25.
- **Research Assistant | Advanced Geometric Computing Lab, IITM** *Jan '25 – Jul '25*
 - Built DGCNN, VGAE and Graph Transformer pipelines to reconstruct midcurves from sparse shape profiles.
 - Benchmarked **10+** methods while studying implicit representations and medial-axis geometry and topology.
 - Reviewed **30+** papers on subdomains in computational geometry, graph learning and geometric deep learning.

SELECTED PROJECTS

- **Multi-view Consistent 3D Reconstruction** *EE5178 Modern Computer Vision; Gaussian Splatting Super Resolution Hackathon*
Mar '26 – Apr '26
 - Achieved **77.2% vs. 56.8%** accuracy (+36%) using 3D Gaussian Splatting on low-resolution multi-view inputs.
 - Replaced COLMAP SfM with HLOC for camera localization, improving the benchmark score from **0.59 to 0.77**.
 - Ablated over **8+** reconstruction pipelines, including SRGS, DAT, HAT, ESRGAN, SwinIR and SplatSure.
 - Improved PSNR from **21 to 29** and SSIM from **75% to 90%+**, preserving fine details in reconstructed views.
 - Evaluated reconstruction quality across challenging scenes focusing on fine-grained text and surface textures.
- **Radar-Based ADAS System** *ED3010 Human Factors in Design*
Aug '25 – Nov '25
 - Built a CARLA/FMCW-radar perception pipeline to assess the risk of collision in fog with **<20 m visibility**.
 - Converted radar detections to Cartesian coordinates in real time and classified threats into **4 hazard levels**.
 - Combined **TTC ≤ 5 s** risk thresholds with radar-HUD alerts and CSV telemetry for real-time decision support.
 - Tested early hazard detection in simulated highway collisions with closing speeds of up to **90 km/h**.
 - Designed a low-cognitive-load HUD and TTC-mapped pulsation to convey hazard urgency without text.

COURSEWORK

Modern Computer Vision; Mathematics for Data Science; Mechatronics System Design; Probability and Statistics; Digital Signal Processing; Control Systems; Data Structures and Algorithms; Human Factors in Design

SKILLS

- **Programming & Tools:** Python, C++, PyTorch, JAX, OpenCV, Git, VSCode, Docker, Linux, Meshlab, Blender
- **3D Vision:** Multi-view Geometry, SfM/HLOC, NeRFs, Gaussian Splatting, Point-Cloud/Mesh Processing
- **Geometric Learning:** SDFs, Implicit Representations, GNNs, Graph Transformers, Shape Reconstruction
- **Representation Learning:** CNNs, U-Nets, Knowledge Distillation, Feature Learning, Data Augmentation

POSITIONS OF RESPONSIBILITY

- **Coordinator | AI Club, IIT Madras** *May '24 – May '25*
 - Organized and Ideated AI Club workshops and events and wrote technical questions as one of **10 coordinators**.
 - Co-organized a workshop on deep learning for **2k+** at CFI Summer School (receiving 40% of total attendees).
 - Taught Gridworld Q-learning to **20+** peers and promoted AI to **1k+** students at CFI Freshie Weekender.
 - Published a Medium review of GameNGen covering reinforcement learning and diffusion-based simulation.